SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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Department of Mechatronics Engineering
23MT507- Prototype Lab

Third Review

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12 RESPONSIBLE CONSUMPTION AND PRODUCTION

Design and Fabrication of Sustainable Smart Water Filling System using Sensor based Automation

6 CLEAN WATER AND SANITATION

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Contents

INTRODUCTION

- In hostels and residential blocks, manually refilling water dispensers from heavy 25-litre cans is tiring and inefficient.
- Delays in refilling often lead to water unavailability during peak hours.
- There is no system to monitor water levels in real time, causing unexpected shortages.
- Manual refilling often leads to spillage and wastage of water.
- Lifting and handling heavy 25-litre cans can cause strain or injuries to staff.

Objectives

The system automates tank filling using sensors to monitor water levels and alert when cans are empty. It keeps the water warm in cold environments through temperature regulation, ensuring consistent availability with minimal manual effort.

Abstract

- Sensor-Based Water Level Monitoring
  → Ensures timely refilling and prevents dispenser from running dry.
- Automated Refill from 25-Litre Cans
  → Reduces manual effort by using a pump to transfer water when levels are low.
- Can Tray Capacity (4 Cans)
  → Allows multiple refills without frequent replacement.
- Empty Can Alert System
  → Notifies admin when all cans are empty, ensuring continuous water availability.
- Temperature Conditioning (Heating Only)
  → Maintains warm water during cold weather for comfort.

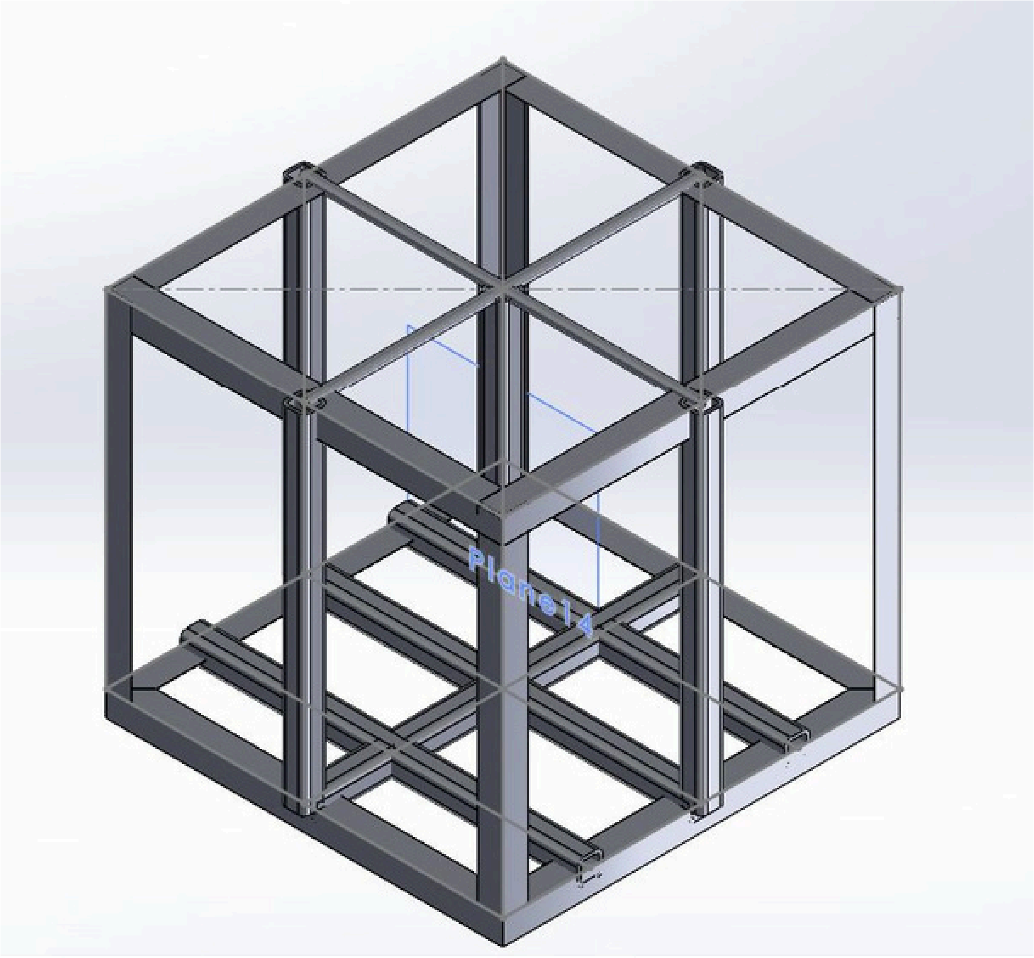
Literature Review

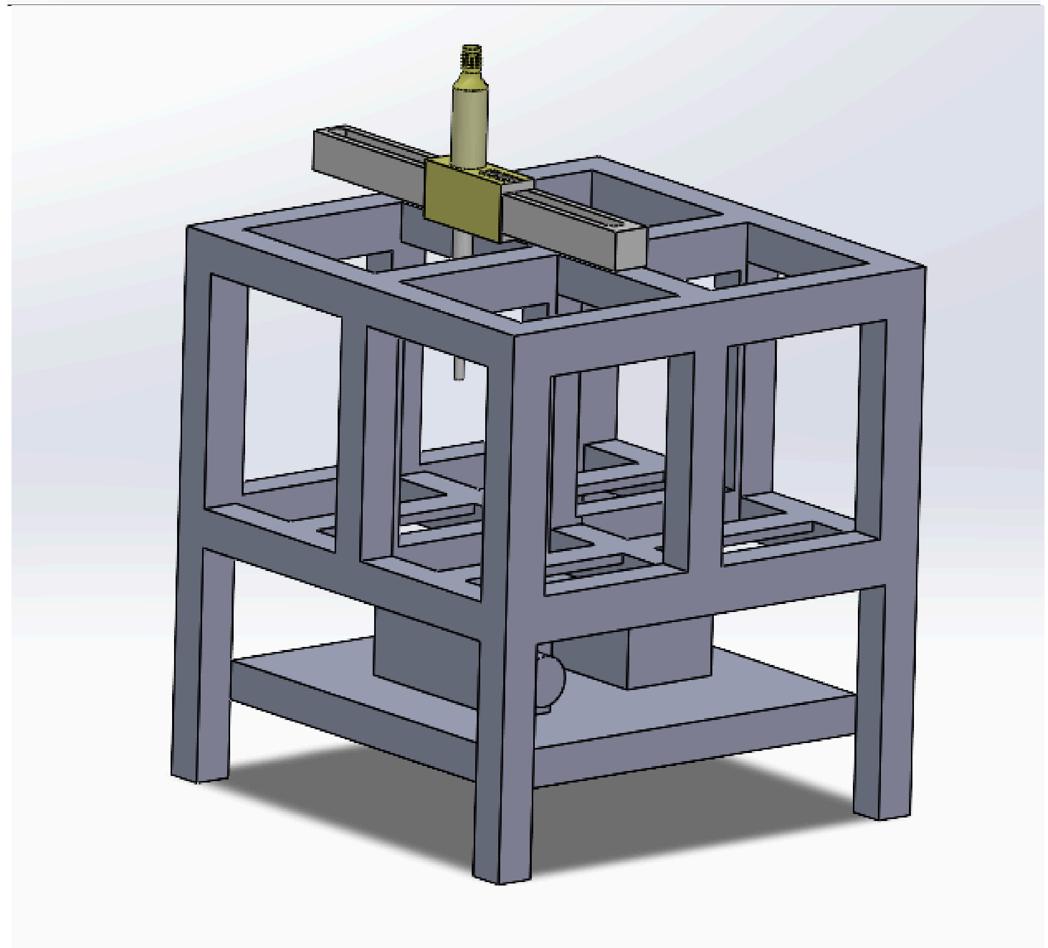
S.No	Author details Year	Concept	Results Observed	Inference Made
1	Mason Dooley and Xiangrong Shenl, 2023	A Novel Self-Actuated Linear Drive for Long-Range-of-Motion Electromechanical Systems	Achieved smooth and precise motion over a 600 mm range with minimal backlash.	belt drives offer cost-effective and silent motion transmission in linear systems.
2	Araujo et al., 2023	Smart mineral water bottle using Internet of Things	Enabled controlled pumping without contamination risk in public dispensers.	Diaphragm pumps provide precise flow control and are ideal for small quantity dispensing systems.
3	L. Kumar & P. Anand, 2022	Designed a 6-finger gripper using servo motors with compliant fingers for adaptive grasping	The gripper was able to pick up irregular and round objects like spheres, bottles, and fruits	Regulated Gripper movement through CAM profile and spring loaded
4	Sandeep Kumar et al., 2021	Designed an automatic water level controller using ultrasonic sensors and Arduino for domestic tanks.	The system successfully monitored and controlled water levels, preventing overflow and dry-run motor issues.	Sensor-based level monitoring reduces water wastage and ensures consistent tank filling without manual effort.
5	Mastura Omar, 2020	Proposed a Smart Water Dispenser Monitoring System using weight sensors and notification modules.	Accurately detected empty cans and sent automated refill alerts to system admin through SMS.	Weight sensors are reliable for tracking water can usage, ensuring timely refills and continuous availability.Using wheatstone bridge

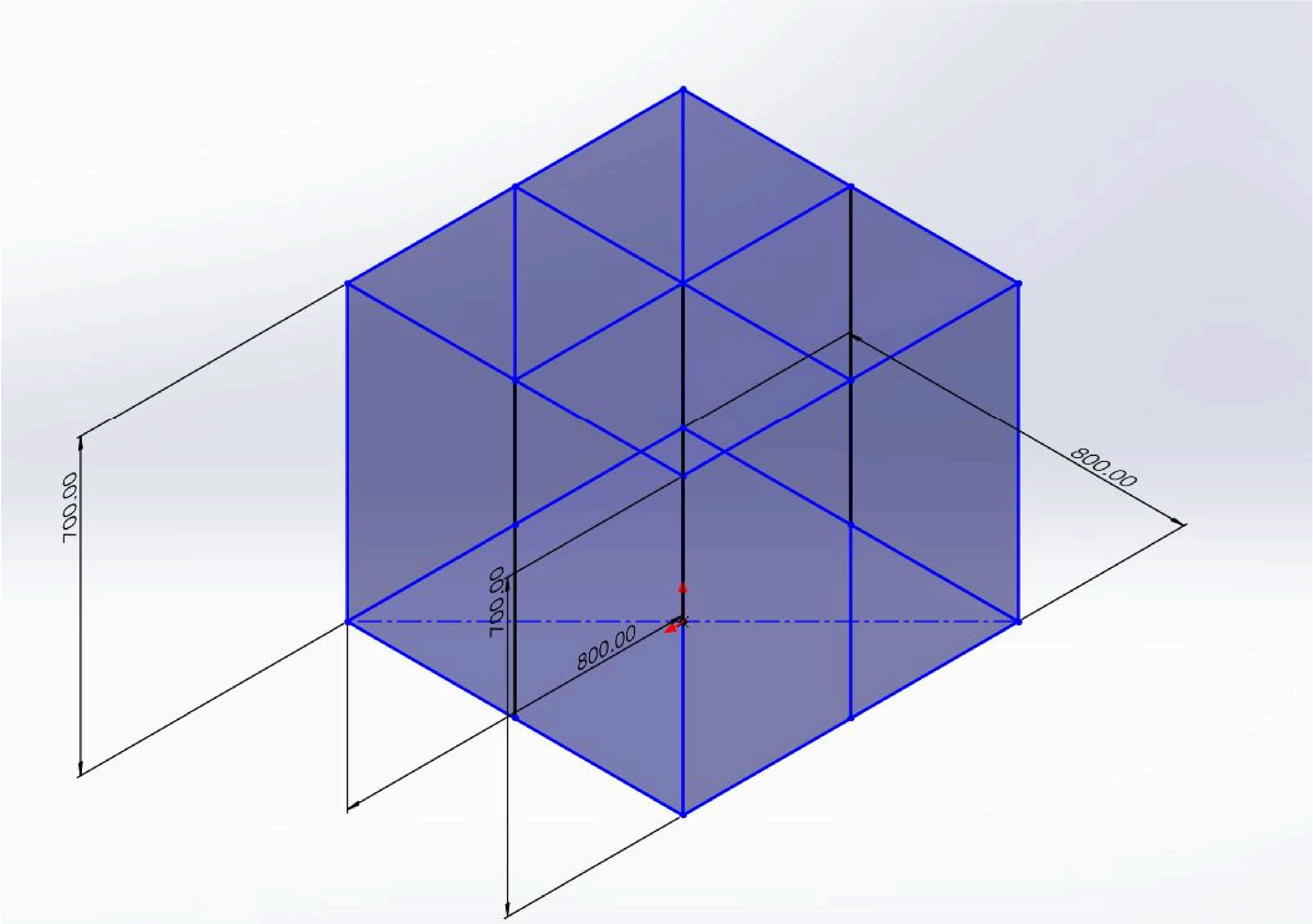
Literature Review

S.No	Author details Year	Concept	Results Observed	Inference Made
6	T. Siva et al., 2020	Integrated a telescopic tube with motor actuation using lead screw mechanism for a pick-and-place robot.	System provided precise control over extension and retraction with ±2 mm repeatability.	Lead screw motor systems provide good control and torque for vertical extension in telescopic tubes.
7	A. Sinha & R. Kumar, 2020	Developed a temperature-controlled water heater using a relay and thermostat system	Maintained water temperature within ±2°C of the set point.	Basic thermostat and relay systems are reliable for automatic heating applications.
8	A. Prakash & M. Ramesh, 2020	Designed a two-finger robotic gripper using an eccentric cam mechanism for actuation.	Achieved synchronized motion of fingers and reliable object grasping with a single motor.	Cam mechanisms simplify motion control by converting rotary motion to linear gripping.
9	R. Kumar & S. Rajan, 2020	Designed a 12V, 5A (60W) flyback converter-based SMPS using TL431 and opto-isolator feedback.	Achieved stable 12V output with ripple <50mV and >80% efficiency.	Flyback topology is ideal for medium-power SMPS with isolation and compact size.
10	R. Balu & S. Kumar, 2020	Used magnetic encoder with DC motor-driven linear actuator for position feedback.	Accurate position control achieved with resolution up to 1 mm.	Encoders allow precise linear motion feedback in real-time.For positioning can neck

3D model







Design Calculation

Bending Moment of Base Frame:

$$M = \frac{WL^2}{8}$$

$$W = \frac{160 \times 9.81}{0.8} = 1962\,\mathrm{N/m}$$

$$M = \frac{1962 \times (0.8)^2}{8}$$

$$= \frac{1962 \times 0.64}{8} = 156.96\,\mathrm{N}$$

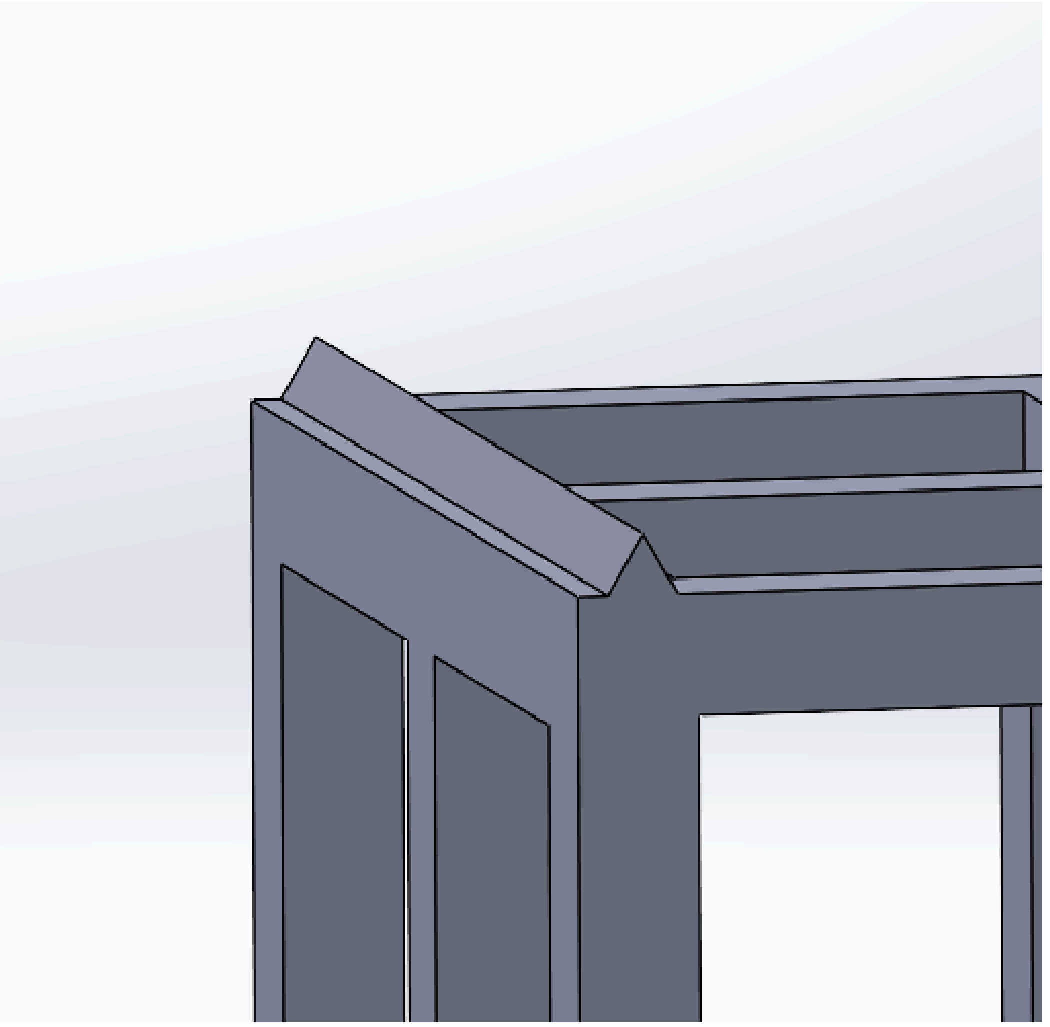
Moment of Inertia of Base Frame:

$$C = 0.04\,\mathrm{m}, \quad I = 3.28 \times 10^{-7}\,\mathrm{m}^4$$

Stress acting on Material:

$$\sigma = \frac{M \cdot C}{I}$$

$$= \frac{6.2784}{3.28 \times 10^{-7}} = 27.8\,\mathrm{MPa}$$



$$\text{Factor of Safety} = \frac{250}{27.8} = 8.9$$

Telescopic tube lead thread selection

- Minor pipe diameter = 19.05 mm
- Major diameter = 24 mm
- **Coarse Series M24**

(Major diameter & minor diameter are choosen based on inner diameter of can neck and pump flow rate)

Telescopic Tube Lead Torque Applied By Motor

Formula : $\tan\alpha = \dfrac{P}{\pi \times D}$

$$\tan\alpha = \frac{3}{\pi \times 22.05}$$

$$\tan\alpha = 0.0433$$

$$\alpha = 2.47^\circ$$

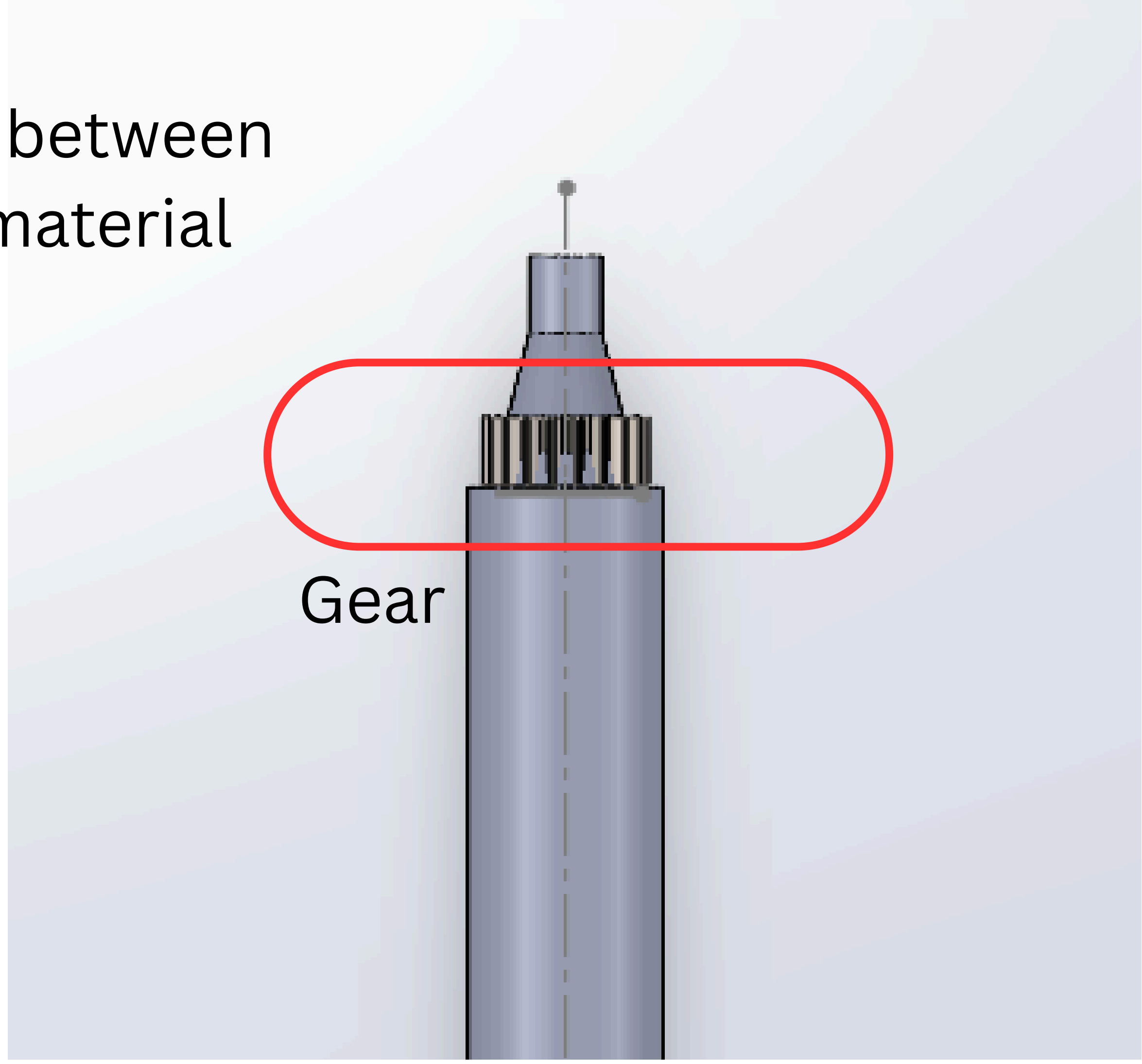
Formula : $p = w \cdot \tan(\pi + \alpha)$

$$P = 250 \times \tan(2.47^\circ + 14.03^\circ)$$

$$P \approx 74.05\,\mathrm{N}$$

Assuming $\mu = 0.25$

= coefficient of friction between threads of mild steel material



$$\tan \Pi = \mu$$

$$\Pi = \tan^{-1} \mu$$

$$\pi = \tan^{-1}(\mu) = \tan^{-1}(0.25) \approx 14.03^{\circ}$$

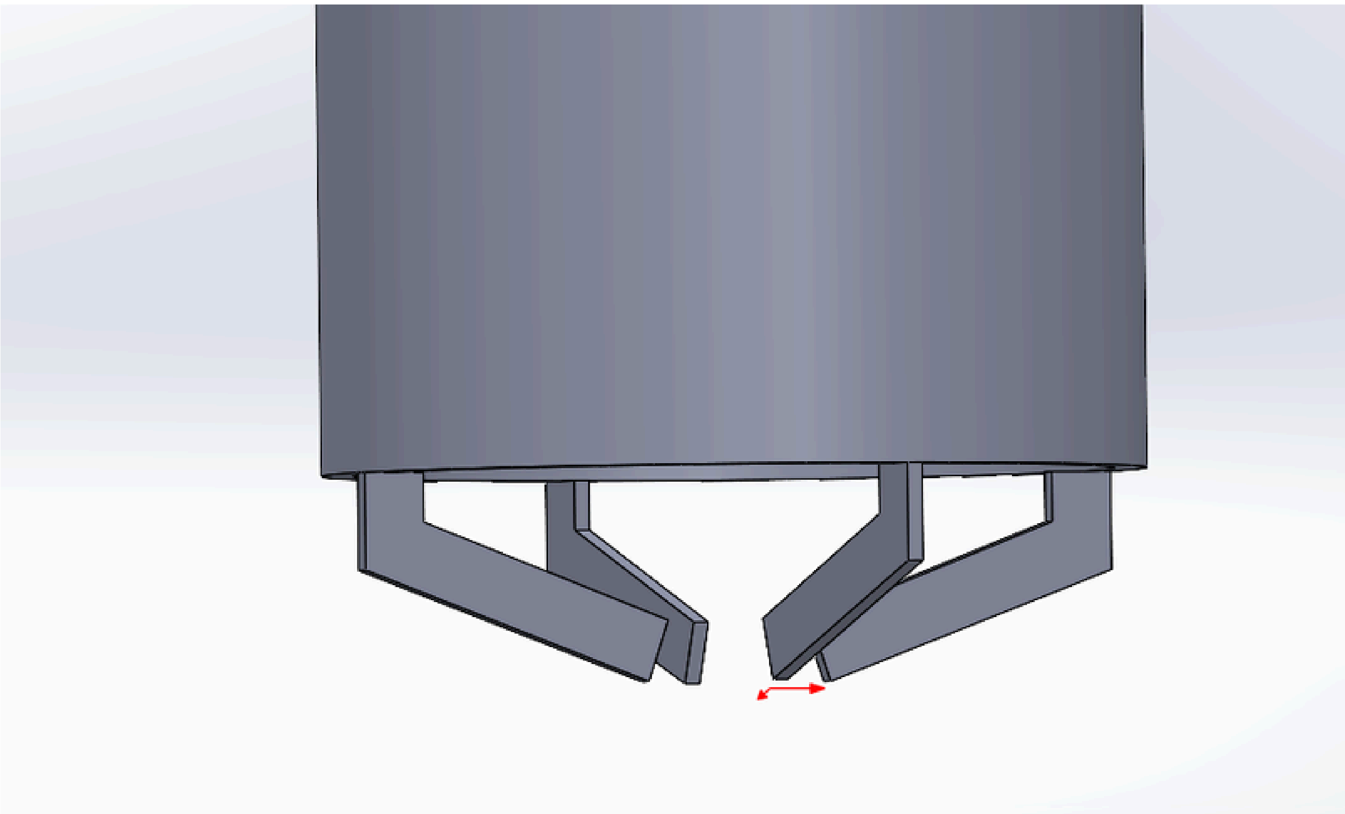
$$\frac{P \times d}{2} \times R = \frac{\text{Gear dia} \times \text{Force}}{2}$$

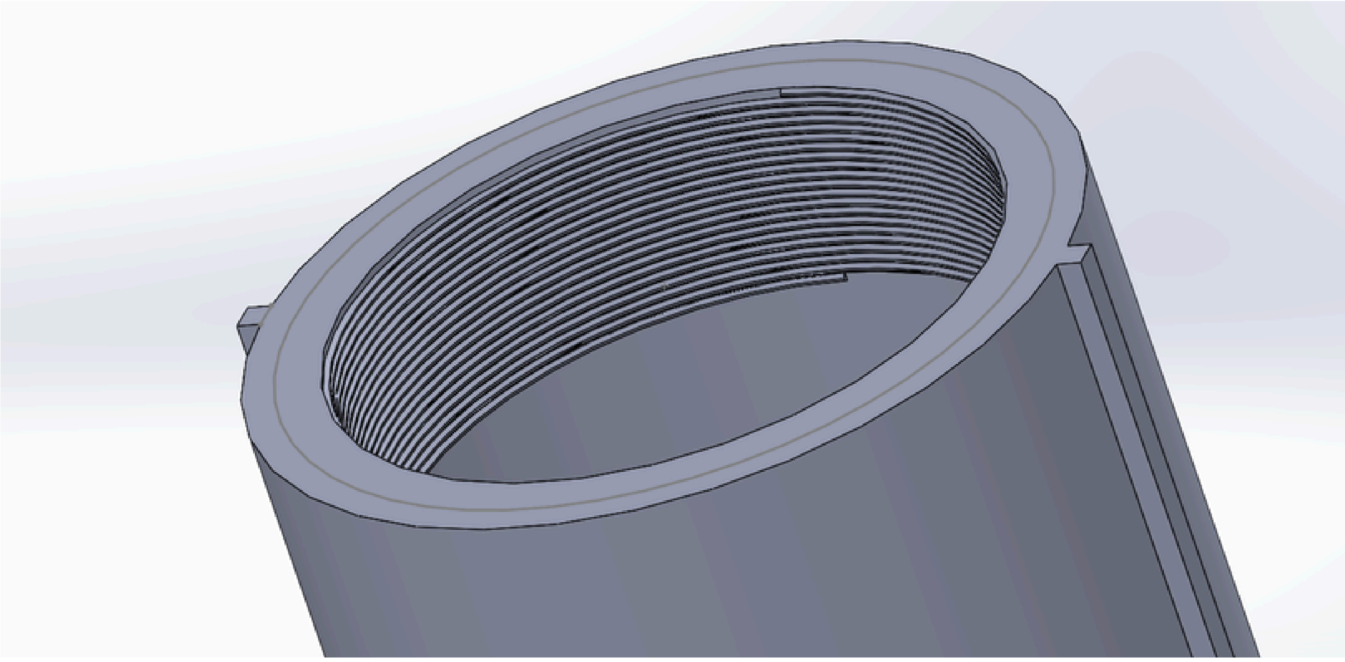
$$\text{Total torque at pipe circumference} = \frac{P \times d}{2}$$

$$\frac{74.05 \times 19.05}{2}$$

$$= \frac{(1348.025 \text{ N-mm})}{2 \times (10^3)}$$

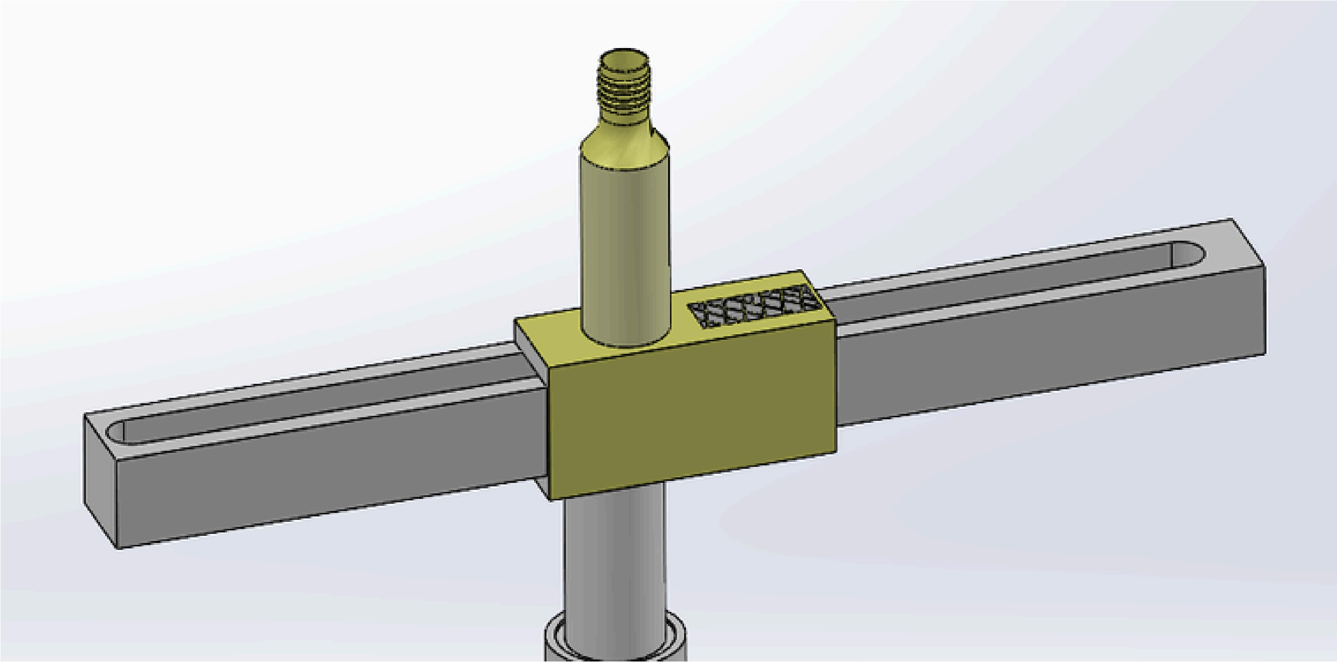
≈0.67 N-m





Gear Calculation :

- Inner diameter (d) - 19mm
- Outer diameter (D) - 38mm
- Torque To transmit = 0.67 n-m
- Pitch/mean diameter (d1) = 28.5mm



Iteration 1 Assuming m = 1.5

$$P_c = \pi \times m$$

$$P_c = \pi \times 1.5 = 4.712\ \mathrm{mm}$$

$$Z = \frac{d_1}{m}$$

$$Z = \frac{28.5}{1.5} = 19$$

$$b = 3 \times P_c$$

$$b = 3 \times 4.712 = 14.137\ \mathrm{mm}$$

Convert d_1 to meters: $28.5 \text{ mm} = 0.0285 \text{ m}$

$$F_a = \frac{T}{d_1/2}$$

$$F_a = \frac{0.67}{0.0285/2} = \frac{0.67}{0.01425} = 47.02 \text{ N}$$

$$Y = 0.154 - \frac{0.912}{Z}$$

$$Y = 0.154 - \frac{0.912}{19} = 0.154 - 0.048 = 0.106$$

Lewis formula:

$$\sigma_b = \frac{F_a}{b \times Y \times P_c}$$

Convert dimensions to meters:

$b = 0.014137 \text{ m}, \ P_c = 0.004712 \text{ m}$

$$\sigma_b = \frac{47.02}{0.014137 \times 0.106 \times 0.004712}$$

$$\sigma_b = \frac{47.02}{7.111 \times 10^{-7}} \approx 6.66 \text{ MPa}$$

Iteration 2

m = 0.8

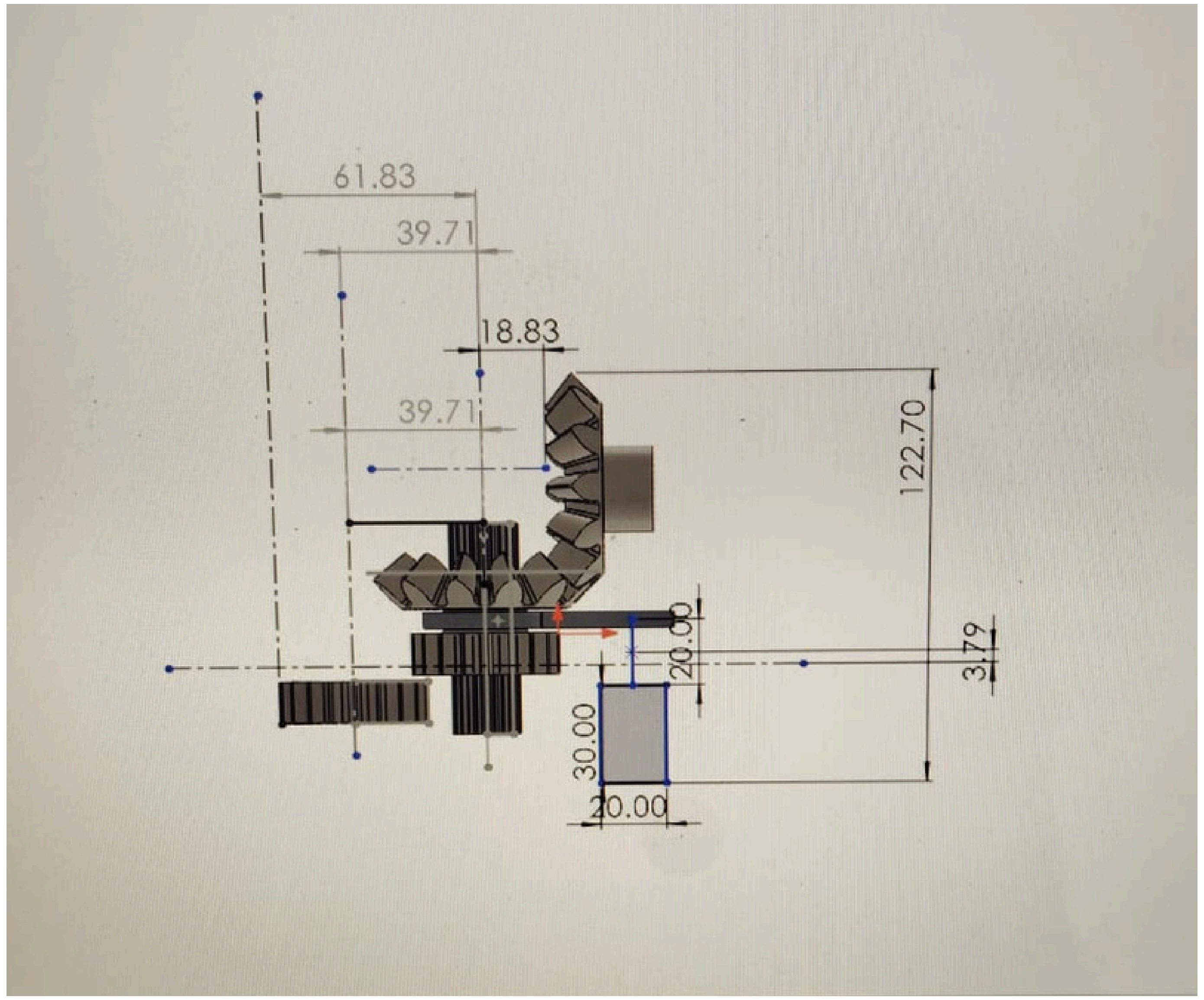
$$\sigma_b = \frac{F_a}{b \times Y \times P_c}$$

With the substituted values shown:

$$\sigma_b = \frac{47}{7.536 \times 10^{-3} \times 0.1284 \times 3.7696 \times 10^{-3}}$$

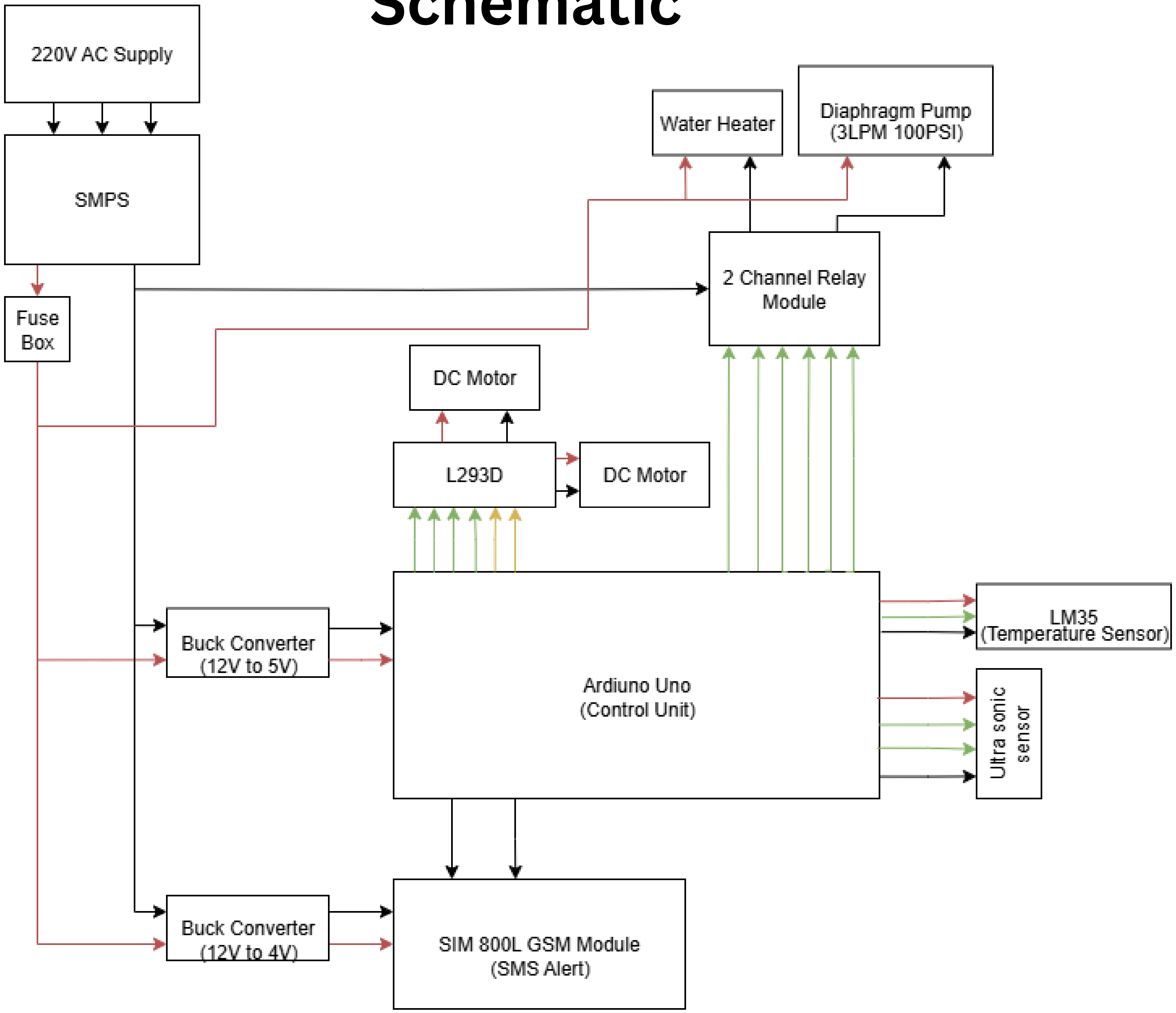
Which gives:

$$\sigma_b = 12.88 \text{ MPa}$$



- **Chosen (Iteration 2):**
  - Module $m = 0.8$ mm
  - Face width $b \approx 7.5$ mm
  - Teeth $Z = 20$ (manufacturer choice / rounded)
  - Inner dia $d_i = 19.5$ mm
  - Addendum dia $A_D = 21.6$ mm
  - Pitch dia $P_D = 20$ mm
- **Calculated bending stress:** $\sigma_b \approx 12.98$ MPa
  - This is **well below** polycarbonate yield (~60–80 MPa), so acceptable.

Schematic



220V AC Supply
SMPS
Fuse Box
Water Heater
Diaphragm Pump (3LPM 100PSI)
2 Channel Relay Module
DC Motor
L293D
DC Motor
LM35 (Temperature Sensor)
Buck Converter (12V to 5V)
Ardiuno Uno (Control Unit)
Ultra sonic sensor
Buck Converter (12V to 4V)
SIM 800L GSM Module (SMS Alert)

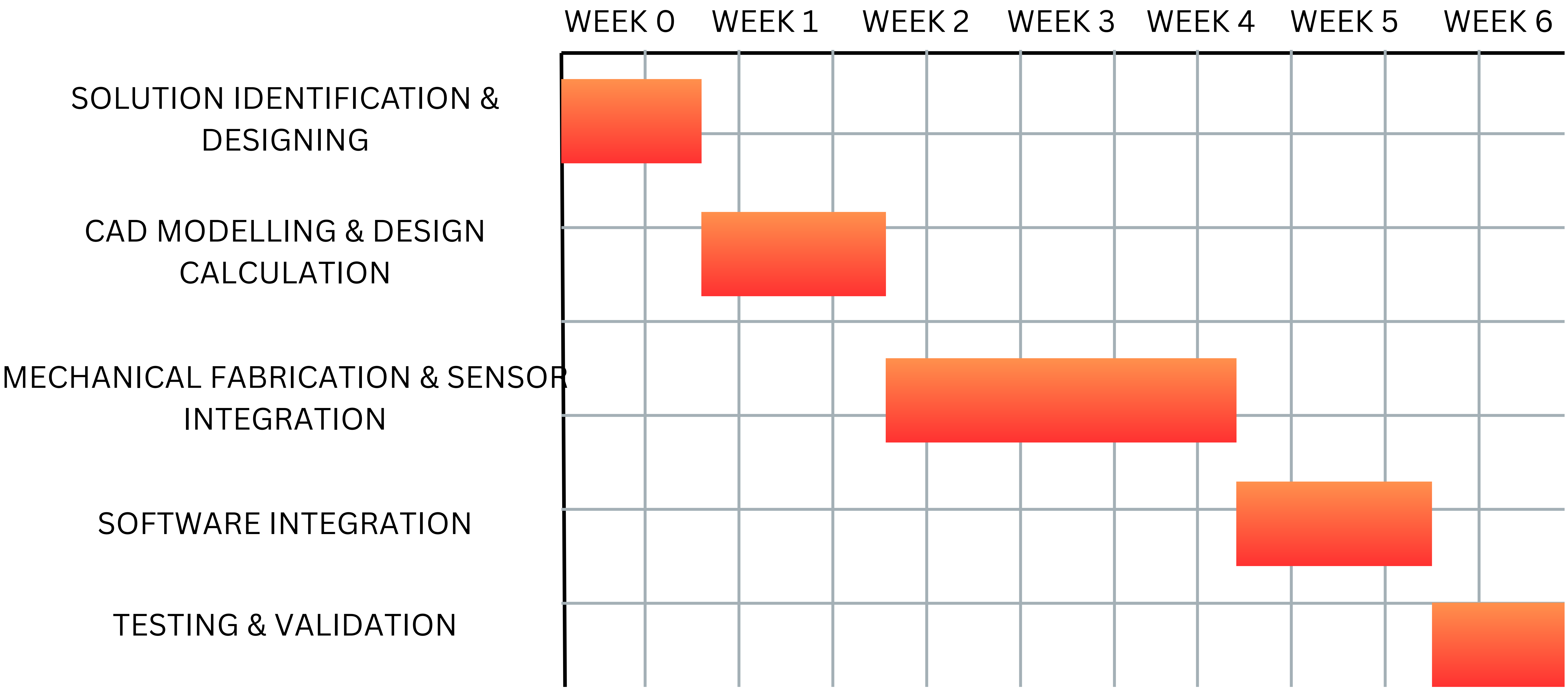
Why Arduino uno?

We need 9 digital pins, RX, TX, GND, and VCC Arduino Uno has all these
It works at 16 MHz, which is enough to drive the L293D motor driver
L293D needs only up to 25 kHz, so Arduino is perfect
It has enough processing power to Control motors, read sensors, handle GSM communication and Control relays.

Why SMPS?

An SMPS (Switched-Mode Power Supply) is recommended because it provides a stable 12V DC output directly from the 230V AC mains. This ensures continuous operation without the need for recharging. SMPS is more economical in the long run due to lower maintenance and no replacement needs, making it ideal for fixed-location, long-duration applications.

GANTT CHART


WEEK 0
WEEK 1
WEEK 2
WEEK 3
WEEK 4
WEEK 5
WEEK 6
SOLUTION IDENTIFICATION & DESIGNING
CAD MODELLING & DESIGN CALCULATION
MECHANICAL FABRICATION & SENSOR INTEGRATION
SOFTWARE INTEGRATION
TESTING & VALIDATION

Bill of Materials

SNo	Components	Specification	Quantity	Price
1	Frame	Mild Steel (MS) or Aluminum, custom size for dispenser tray	1 Set	₹1700
2	Reservoir Tank	Food-grade plastic or stainless steel, 5–10 litres buffer (upto 40°C)	1	₹250
3	3/4 inch Hose	Food-grade PVC, 3/4 inch diameter, 10 feet length	10 feet	₹200
4	ATmega 328p	Microcontroller (Arduino Uno based)	1	₹400
5	Servo Motor	12V, 10–20 kg-cm torque (for cap removal & mechanism)	1	₹1000
6	SMPS Unit	12V, 5A DC regulated power supply	1	₹1000
7	Diaphragm Water Pump 12V Self-Priming	12V, 3–4 LPM flow, 100 PSI suction pressure	1	₹400
8	LM35 Temperature Sensor	Digital temperature sensor (-55°C to +125°C)	1	₹150
9	Ultrasonic Sensor	HC-SR04, 2cm–400cm detection range	1	₹50

SNo	Components	Specification	Quantity	Price
10	Telescopic Tube	Aluminum, 2-stage, 25mm–50mm diameter, ~1m extendable length	1	₹200
11	Stepper Motor	NEMA 17, 1.8° step angle, 12V, 1.5A, 40–45 N·cm holding torque (with driver module)	2	₹400
12	Open Timing Belt (6mm Width)	GT2 type, rubber with fiberglass core, 6mm width, 3m length	3m	₹500
13	Nylon Gear	30–40mm diameter, 1:2 ratio, bore size compatible with actuator shafts	6	₹400
14	GSM Module (SIM800)	GSM (supports SMS alerts)	1	₹400
15	Coil Type Heater	1000–1500W, 230V AC, Stainless Steel immersion coil, suitable for 5–10 litre tank	1	₹500

Estimated Total Prototype Cost

₹7,550 (approx.)

References

1) V.B. Bhandari, Design of Machine Elements, McGraw-Hill Education.
Provides fundamentals of designing mechanical parts such as shafts, couplings, and actuators used in the dispenser frame and pump mechanism.

2) J.L. Meriam and L.G. Kraige, Engineering Mechanics: Dynamics, Wiley. Covers force and motion analysis, essential for understanding motor-driven actuation of telescopic tubes and the handling of water cans.

3) Max Rabiee, Programmable Logic Controllers (PLCs): Hardware and Programming,Cengage Learning.
Explains the basics of automation and control, relevant for microcontroller/PLC-based water level monitoring and pump actuation.

4) Mikell P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, Pearson.
Provides insights into integrating automation, robotics, and control systems into real-world applications like automated dispensing.

5) William D. Callister and David G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley.
Useful for selecting food-grade, corrosion-resistant materials for water storage tanks and dispenser frames.

THANK

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troduction

Literature Review

Design Calculation

Architecture Design

Components of water filling system

cost estimation

advantages and disadvantages

applications

References